

Jeffrey Bratman

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SUMMARY

Aerospace engineer with hands-on experience designing, integrating, and testing UAVs and autonomous racing vehicles from concept through flight and track validation. Skilled in CAD-driven design (Fusion, SolidWorks, NX), structural analysis, and manufacturing with composites, additive, and sheet-metal methods. Experienced in ROS2-based autonomy integration, embedded hardware/software validation, and data-driven performance testing. Combines aerospace precision with hands-on motorsports engineering to deliver robust, high-efficiency mechanical and system-level solutions.

EDUCATION

University of California-San Diego

Bachelor of Science in Aerospace Engineering

Sept. 2022 – June 2025

Mira Costa Community College

Associate in Arts in Liberal Arts Emphasis in Mathematics and Sciences

Aug. 2020

Certificate of Achievement in Automotive Technology

Dec. 2012

SKILLS

- **CAD Software:** Autodesk Fusion, SolidWorks, Siemens NX, AutoCAD, Creo
- **Manufacturing & Prototyping:** Additive Manufacturing (FDM, SLA, SLS), Composites, sheet metal, machining
- **Programming & Analysis:** Python, MATLAB, C++ (in progress); MoTeC i2, ChassisSim, FEA (Coursework)
- **Autonomy:** ROS2, Docker, VMware, Linux (Embedded & Systems)

EXPERIENCE

UC San Diego Autonomous Vehicles Racing Team (UCSD)

Aug. 2025 – Present

Post-graduation Research Assistant

- Redesigned sensor mounts for autonomous racecars, improving instrumentation reliability and data accuracy.
- Developed and executed integration and test plans for vehicle subsystems.
- Improved accessibility and serviceability of subsystems, reducing downtime and enabling rapid test turnaround.

Triton AI – Indy Autonomous Challenge (UCSD)

Nov. 2024 – Aug. 2025

Vehicle Dynamics Engineer

- Developed a vehicle chassis simulation model (ChassisSim + MoTeC i2) to predict dynamics and optimize performance.
- Performed telemetry-based validation and parameter tuning to enhance handling and stability.
- Coordinated between mechanical and controls teams for iterative development cycles.

Triton AI – Autonomous Karting Series (UCSD)

Nov. 2024 – Aug. 2025

Head of Mechanical Development

- Designed electrical and battery mounting systems to improve thermal management and structural integrity.
- Implemented modular subsystems enabling rapid configuration changes for race testing.
- Coordinated with project leads to ensure timely system integration.

N.A.S.A L'Space Proposal Writing & Evaluation Professional Development Program

May. 2024 – Aug. 2024

Head of Programmatic Team

- Co-authored a proposal addressing a NASA mission pain point, including supporting intellectual property.
- Led logistics, budgeting, and communications between PI and technical team.
- Managed deliverables to ensure technically and financially feasible submission.

Void Putters (San Diego)

Aug. 2022 – Sep. 2022

Engineering Consultant

- Designed testing procedures to evaluate prototype performance and validate new product technology.
- Proposed and implemented materials integration strategies to improve product function and manufacturability.

101 European (San Diego)

Jan. 2019 – Mar. 2020

Assistant Shop Manager

- Supervised daily shop operations and led troubleshooting/fabrication for European performance vehicles (BMW, Audi, Porsche).